

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	Date:	

Code of Conduct

I G. Jaswanth - 192472235 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Jaswanth

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	4	3	3	18	94
2	4	4	4	3	3	18	
3	4	3	3	4	4	18	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Reference Resolution and Discourse Coherence

Given Paragraph

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	Book on Artificial Intelligence
they	Ravi and Arun
the book	Book on Artificial Intelligence

b) Constraint-Based Reference Resolution

The references are resolved by applying the given constraints: **gender and number agreement**, **recency**, **grammatical role**, **semantic compatibility**, and **discourse coherence**.

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Ravi is singular and male; grammatically compatible with “He”.
He	Arun	Valid	Arun is also singular and male; therefore gender/number alone cannot distinguish them.
him	Ravi	Valid	Ravi is singular male and can be the object of “helped”.
him	Arun	Invalid	Arun is the subject performing the action “helped”, so referring to Arun as “him” creates an inconsistent role in the sentence.
it	Book	Valid	“It” is singular and semantically refers to the object being found.
it	Ravi/Arun	Invalid	People are not compatible with the object reference in “find it”.
they	Ravi + Arun	Valid	“They” is plural and refers naturally to the two previously introduced people.

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
the book	Book on Artificial Intelligence	Valid	The previously mentioned book is the only suitable discourse entity.

Constraint Analysis

1. Gender and Number Agreement:

“Ravi” and “Arun” are both singular male entities, so both can initially satisfy “**He**”. “They” requires a plural antecedent, making Ravi + Arun the appropriate interpretation.

2. Recency:

“Arun” is the most recently mentioned person before “He”. Therefore, recency favors **Arun**.

3. Grammatical Role:

In “Ravi met Arun”, Ravi is the subject and Arun is the object. The following sentence does not provide enough grammatical evidence to completely distinguish the two male candidates.

4. Semantic Compatibility:

In “Arun helped him find it”, “**it**” must refer to the book because a book is something that can be found. “Him” refers to the person receiving help.

5. Discourse Coherence:

The paragraph is most coherent when Ravi is the person looking for the book, while Arun helps him find it. This gives the sequence: **Ravi searches** → **Arun helps** → **both discuss the book**.

Final Resolution

- **He** → Ravi
- **him** → Ravi
- **it** → the book
- **they** → Ravi and Arun
- **the book** → the previously mentioned book on Artificial Intelligence

c) Final Reference Resolution Table

Referring Expression	Resolved Antecedent	Main Constraint
He	Ravi	Discourse coherence
him	Ravi	Grammatical role + coherence
it	Book on Artificial Intelligence	Semantic compatibility
they	Ravi and Arun	Number + discourse coherence
the book	Book on Artificial Intelligence	Discourse continuity

d) Final Coreference Chains

The coreference chains are:

Person Chain 1:

Ravi → He → him → they

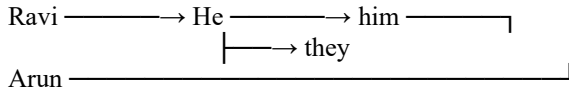
Person Chain 2:

Arun → they

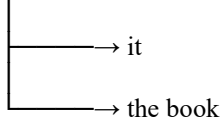
Object Chain:

a book on Artificial Intelligence → it → the book

Coreference Representation



a book on Artificial Intelligence



e) Paragraph with References Replaced

“Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.”

This version removes the ambiguity caused by the pronouns and makes all entity relationships explicit.

f) Ambiguity if “He” Appeared Before Arun Was Introduced

If the paragraph were:

“Ravi met someone at the library. He was looking for a book on Artificial Intelligence. Arun was introduced later.”

the system would initially have less information about the antecedent of “He”. If Arun had not yet been introduced, **Arun could not be considered an antecedent at that point.**

More generally, when multiple singular male entities have already been introduced, “He” becomes **ambiguous** because both satisfy gender and number agreement. The system must therefore use additional constraints such as **recency, grammatical role, semantic compatibility, and discourse coherence** rather than relying only on gender and number.

Conclusion

Constraint-based reference resolution is necessary because a pronoun may have multiple grammatically possible antecedents. In this paragraph, combining the constraints allows the system to establish a coherent interpretation: **Ravi is searching for the book, Arun helps Ravi find it, and both later discuss the book.** This demonstrates how constraint satisfaction maintains consistent entity chains throughout a discourse.

Q2. Dialog Act Recognition and Response Generation

Given Conversation

Student: “I submitted my assignment, but I am worried that I may have made many mistakes.”

Required Dialog Acts

Reassure + Advise

a) Identify the Dialog Acts Required

The response must perform two dialog acts:

1. **Reassure** – Reduce the student's worry and provide confidence.
2. **Advise** – Suggest a constructive action, such as reviewing the assignment and using feedback to improve.

The response should perform both acts naturally without contradicting the student's concern.

b) Important Entities to Maintain

The following entities must remain consistent throughout the response:

Entity	Role
Assignment	The work submitted by the student
Mistakes	The student's concern about errors
You	Refers consistently to the student

The response should remain focused on the submitted **assignment** and the possibility of **mistakes**, rather than introducing unrelated topics.

c) Three Possible Responses

Response 1

“It is completely understandable to feel worried after submitting an assignment, but you should remain **confident** about your effort. You can **review** the feedback when it is available and use it to **improve** your future assignments.”

Response 2

“Don't worry too much about the **mistakes**; submitting your assignment is an important step. **Review** the feedback when you receive it so you can **improve** your work and feel more **confident** next time.”

Response 3

“It is normal to be concerned about possible **mistakes**, so try to stay **confident** about the assignment you submitted. Use the **feedback** to **improve** your understanding and performance in future assignments.”

All three responses:

- Provide reassurance.
- Provide advice.
- Maintain entity coherence.
- Use at least two required keywords.
- Use 2 sentences.
- Maintain a positive and polite tone.
- Do not provide contradictory advice.

d) Comparison of Responses

Criteria	Response 1	Response 2	Response 3
Coherence	High	High	High
Constraint Satisfaction	High	High	High
Politeness	High	High	High
Relevance	High	High	High
Overall	Very Good	Excellent	Very Good

Analysis

Response 1:

Clearly connects the student's concern with reviewing feedback. It provides both reassurance and practical advice.

Response 2:

Directly addresses the student's **mistakes**, provides reassurance, and then gives a clear action: review feedback and improve future work. It maintains a strong cause-effect relationship.

Response 3:

Provides a supportive response and connects feedback with future improvement. It is relevant but slightly more general than Response 2.

e) Best Response and Justification

Selected Response: Response 2

“Don't worry too much about the mistakes; submitting your assignment is an important step. Review the feedback when you receive it so you can improve your work and feel more confident next time.”

Justification

Response 2 is the best because it satisfies **all the specified constraints**:

- **Reassurance:** “Don't worry too much...”
- **Advice:** “Review the feedback...”
- **Entity coherence:** Maintains the assignment and mistakes throughout.
- **Required keywords:** Uses **review, improve, feedback, and confident**.
- **Discourse relation:** Establishes a clear **cause-effect relationship**:
 - Review feedback → improve work → become more confident.
- **Length:** Exactly **2 sentences**.
- **Tone:** Positive, supportive, and polite.

- **Relevance:** Directly addresses the student's concern.

Therefore, Response 2 provides the strongest balance of **coherence, constraint satisfaction, politeness, and relevance**.

f) Effect of Not Maintaining Entity Coherence

If entity coherence is not maintained, the generated response may become confusing or irrelevant.

For example:

“Don't worry about your assignment. The project can be improved by reviewing the weather forecast.”

Here, the response shifts from the **assignment** to an unrelated **weather forecast**. The entities and topic are no longer consistent with the conversation.

This can result in:

- Loss of conversational context.
- Confusing or irrelevant responses.
- Reduced user trust.
- Incorrect interpretation of the user's concern.
- Poor-quality dialogue generation.

Conclusion

Constraint-based dialog generation ensures that the response remains **contextually relevant, coherent, polite, and consistent** with the user's conversation. Applying dialog acts, entity constraints, discourse relations, keyword constraints, and response-length requirements produces a reliable conversational response.

Q3. Word Sense Disambiguation and Meaning Representation

Given Sentence

“Priya sat on the bank and watched the boats moving across the water.”

The word “**bank**” is ambiguous because it can have two different meanings:

1. **Financial institution**
2. **Side of a river**

a) Possible Meanings of “Bank”

Sense	Meaning	Example
Sense 1	Financial institution	Priya went to the bank to deposit money.
Sense 2	Side of a river	Priya sat on the bank and watched the boats.

b) Correct Sense and Contextual Constraints

Correct Sense: Sense 2 – Side of a River

The correct interpretation of “**bank**” is **the side of a river or water body**.

Reasoning

The surrounding words strongly indicate the intended meaning:

- “**sat on the bank**” → a person can physically sit on the side of a river.
- “**watched the boats**” → boats are associated with water bodies.
- “**moving across the water**” → provides a strong contextual connection between the bank and water.
- There is no financial context such as **money, account, deposit, loan, or transaction**.

Therefore:

bank → **river bank**

This demonstrates how **contextual constraints** can resolve lexical ambiguity.

c) Predicate Logic Representation

Let:

- **Priya** = pp
- **Bank** = bb
- **Boats** = zz
- **Water** = ww

Using the predicates provided:

- $\text{sit}(x)\text{sit}(x)$
- $\text{bank}(y)\text{bank}(y)$
- $\text{beside}(x,y)\text{beside}(x,y)$
- $\text{watch}(x,z)\text{watch}(x,z)$
- $\text{boat}(z)\text{boat}(z)$
- $\text{move}(z)\text{move}(z)$
- $\text{water}(w)\text{water}(w)$

A suitable representation is:

$\text{sit}(\text{Priya}, \text{bank})$
 $\text{bank}(\text{bank})$
 $\text{beside}(\text{Priya}, \text{bank})$
 $\text{watch}(\text{Priya}, \text{boat})$
 $\text{boat}(\text{boat})$
 $\text{move}(\text{boat})$
 $\text{water}(\text{water})$

Structured Predicate Logic Representation

$\exists b, z, w [\text{bank}(b) \wedge \text{water}(w) \wedge \text{beside}(\text{Priya}, b) \wedge \text{sit}(\text{Priya}, b) \wedge \text{boat}(z) \wedge \text{move}(z, w) \wedge \text{watch}(\text{Priya}, z)]$

Meaning

Priya is sitting beside a river bank, the bank is associated with water, and Priya is watching boats that are moving across the water.

d) Unambiguous English Paraphrase

“Priya sat on the side of the river and watched the boats moving across the water.”

This paraphrase explicitly replaces the ambiguous word **“bank”** with **“side of the river”**, removing the possibility of interpreting it as a financial institution.

e) How WSD Improves Machine Translation

Word Sense Disambiguation (WSD) helps a machine translation system determine the correct meaning of an ambiguous word before translating it.

For example:

“Priya sat on the bank and watched the boats.”

If the system incorrectly selects the **financial institution** sense of “bank”, it may produce an incorrect translation referring to a financial organization.

By using contextual information such as **boats**, **water**, **sat on**, and **moving**, the system selects the **river-bank sense**.

Therefore, WSD:

- Reduces incorrect word translations.
- Preserves the intended meaning of the source sentence.
- Improves semantic accuracy.
- Produces more coherent translations.
- Helps translation systems handle ambiguous words according to their context.

Conclusion

The word **“bank”** has two possible senses, but the contextual clues **“boats”** and **“water”** strongly support the **river-bank** interpretation. Constraint-based WSD combined with predicate logic preserves the semantic relationships and enables the system to generate an unambiguous and meaningful interpretation.